

Light It Up

TODAY YOU'LL MAKE

A light of your own that turns on and off.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ USB cable
- ☐ LED
- ☐ 2 alligator clips
- ☐ Pencil

STEPS

- 1 Clip a wire from pin **0** to the LED's **long** leg.
- 2 Clip a second wire from **GND** to the LED's **short** leg.
- 3 In MakeCode, click **Advanced**, then **Pins**.
- 4 On button A pressed, add **digital write pin P0 to 1**. Button B, make it **0**.
- 5 Download it. Press A. Press B.

IF IT BREAKS

LED never lights? Turn it around. Long leg to P0, short leg to GND.

Very dim? Push the clips onto the bare metal legs, not the plastic.



SCAN ME

YOUR CHALLENGE

Make the light blink on its own — on, pause, off, pause, inside a **forever** loop.

DRAW YOUR CIRCUIT

**I DID IT!**

Date: _____

ONE THING I LEARNED

Make Some Noise

TODAY YOU'LL MAKE

A micro:bit that plays a tune out loud.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ USB cable
- ☐ Buzzer
- ☐ 2 alligator clips
- ☐ Pencil

STEPS

- 1 Clip a wire from pin **0** to one leg of the buzzer.
- 2 Clip a second wire from **GND** to the buzzer's other leg.
- 3 Click **Music**. Drag **start melody** into **on button A pressed**.
- 4 Swap it for **play tone Middle C for 1 beat**. Try other notes.
- 5 Download it. Press A.

IF IT BREAKS

No sound at all? Swap the two clips around, then check both grip bare metal.

Too quiet? Hold the buzzer flat against the table. It gets much louder.



SCAN ME

YOUR CHALLENGE

Play the first line of Happy Birthday. Write the notes down before you build it.

WRITE YOUR TUNE OUT

**I DID IT!**

Date: _____

ONE THING I LEARNED

The Breadboard

TODAY YOU'LL MAKE

A traffic light that runs by itself.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ Edge connector
- ☐ Breadboard
- ☐ 3 LEDs
- ☐ 3 resistors
- ☐ Jumper wires

STEPS

- 1** Push the micro:bit into the **edge connector**. Plug that into the breadboard.
- 2** Put a red, a yellow and a green LED in three **separate** rows.
- 3** Run a **220Ω resistor** from each short leg to the blue – rail.
- 4** Run a jumper from **P0, P1 and P2** to each long leg.
- 5** Light them in order with **digital write** and **pause**. Download it.

IF IT BREAKS

Only one LED works? Each LED needs its own row. Two legs in one row is a short cut.

Nothing lights at all? Run one jumper from **GND** to the blue – rail.



SCAN ME

YOUR CHALLENGE

Time a real traffic light on your street. Change your pauses until yours matches.

DRAW THE BOARD AND THE WIRES

**I DID IT!**

Date: _____

ONE THING I LEARNED

Make It Move

TODAY YOU'LL MAKE

An arm that waves when you press a button.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ Edge connector
- ☐ Servo
- ☐ Jumper wires
- ☐ Cardboard + tape

STEPS

- 1** Plug the servo's **brown** wire to GND, **red** to 3V, **orange** to P0.
- 2** Click **Advanced**, then **Pins**. Find **servo write pin P0 to 180**.
- 3** Put it inside **on button A pressed**.
- 4** Add another on button B. Set that one to **0**.
- 5** Download it. Tape a cardboard arm to the servo. Press A.

IF IT BREAKS

Buzzes and shakes? It's pushing past its end stop. Stay between 0 and 180.

Doesn't move? Check the red wire is on **3V**, not GND. The colours matter.



SCAN ME

YOUR CHALLENGE

Make it wave on its own — 0, pause, 180, pause, forever. Then give it a face.

DRAW THE ARM YOU'LL TAPE ON

**I DID IT!**

Date: _____

ONE THING I LEARNED

Is the Plant Thirsty?

TODAY YOU'LL MAKE

A sensor that tells you when to water a plant.

GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ 2 nails
- ☐ 2 alligator clips
- ☐ A plant
- ☐ Pencil

STEPS

- 1** Clip a wire from **P1** to one nail, and **GND** to the other nail.
- 2** Push both nails into the soil, a finger's width apart.
- 3** Put **show number** → **analog read pin P1** inside **forever**.
- 4** Write down the number dry. Water the plant. Write it down wet.
- 5** Add an **if**: below your dry number, show a sad face.

IF IT BREAKS

Number jumps around? Normal. Pick a number halfway between your dry and wet readings.

Always says 0? A nail has come loose. Check both clips grip bare metal.



SCAN ME

YOUR CHALLENGE

Add the buzzer from Week 12. Make it beep until someone waters the plant.

WRITE YOUR DRY AND WET NUMBERS

**I DID IT!**

Date: _____

ONE THING I LEARNED
